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Advanced Data Communication: Principles and Applications of Satellite Communication

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List of Abbreviations

ACM – Adaptive Coding and Modulation

The method of removing debris that is already activated. Active Debris Removal.

APSK is a modulation technique of amplitude and phase. APSK: Amplitude and phase modulation

ASAT – Anti - Satellite

CAEPX – Capital Expenditure

DVB-S2X is an enhancement of the Digital Video Broadcasting – Satellite (DVB-S2) standard.

FEC – Forward Error Correction: This is a forward error correction method for audio.

FSS – Fixed Satellite Service

They are of the type: Geostationary Earth Orbit (GEO).

GNSS - Global Navigation Satellite System

HAPS – High-Altitude Pseudo-Satellite

HTS – High-Throughput Satellite

IFC – In-Flight Connectivity

IoT – Internet of Things

ISL – Inter-Satellite Link

ISR – Intelligence, Surveillance and Reconnaissance

ITU – International Telecommunication Union

LEO – Low Earth Orbit

Long Fat Night.

LDPC – Low Density Parity-Check

MEO – Medium Earth Orbit

MNO – Owners of the mobile network.

The key is to use Non-Orthogonal Multiple Access (NOMA).

NR – New Radio

NTN – Non-Terrestrial Network

OPEX – Operational Expenditure

PAPR – Peak-to-Average Power Ratio

PMD – Post-Mission Disposal

RF – Radio Frequency

RTT – Round-Trip Time

SAR – Synthetic Aperture Radar

SDR – Software-Defined Radio

SNR – Signal-to-Noise Ratio

Space Situational Awareness (SSA)

SSPA – Solid-State Power Amplifier

The protocol is TCP – Transmission Control Protocol.

THz – Terahertz

TWTA: Travelling wave Tube Amplifier.

A VSAT is a transmission/reception station that is small in size.

Abstract

The modern digital infrastructure rests on the ability to quickly, rapidly and universally exchange data. The terrestrial fiber-optic network is at the heart of the urban communication infrastructure, whereas satellite data communication is an unreplaceable paradigm for global coverage, especially for maritime, aeronautical and remote terrestrial applications. The complexity and functionality of contemporary satellite systems, ranging from the basic geostationary relay to the complex multi-orbit constellation, will be addressed in this research paper. The technical discussion of the “Satellite Zooming” concept, which is enabled by HTS (High Throughput Satellites) using high gain spot beam technology, and aggressive frequency reuse is crucial to this analysis. The discussion is structured to deal with the fundamentals of orbital mechanics in the Low Earth Orbit (LEO), Medium Earth Orbit (MEO) and Geostationary Orbit (GEO) as well as high level frequency spectrum management in the C, Ku, Ka and upcoming V/W bands. In addition, more sophisticated signal modulation techniques are presented and analyzed in the essay, which are essential for realizing maximum spectral

efficiency. Satellite versus terrestrial communications is presented and compared in detail, including a discussion of the compromises involved, such as latency, bandwidth and deployment costs. The story then comes to some critical issues including atmospheric attenuation and propagation delay, growing threat from orbital debris, and deep cyber security weaknesses. Lastly, the essay looks forward and identifies the key role that satellite data communication will play in the future 6G networks and how AI will help in the autonomous operation of the satellites and dynamic allocation of resources.

1. Introduction: Evolution of Satellite Communication

Satellites have come a long way from their early days to become an integral part of the global data infrastructure today. Satellite communication has traveled a long way since the early days and is now an integral part of today's global data fabric.

1.1. History and Origins of SBDE

The first generation of satellite technology was conceived to span vast distances, mostly used as a large transponder for analog voice and narrowband TV. These early systems were 'bent-pipe' since they took in a signal from a ground station, changed its frequency to one where the uplink would not interfere and boosted the signal for retransmission to Earth. There was no onboard processing, no onboard intelligence routing and not much thought was given to the spectral efficiency. The bandwidth was very narrow and the ground infrastructure was huge to cope with the feeble signals coming from space.

1.2. Satellites' role in the modern global data infrastructure

Now, however, the explosive development of the Internet, the proliferation of the digital economy and society's ever-growing demands for ubiquitous connectivity have changed the architectural and functional needs of satellite data communication to a degree that is

unprecedented. Satellite communication is no longer the accessory of terrestrial communication, but one of the important links and independent layers in the modern global data infrastructure in order to achieve seamless global coverage. Fiber optics have the most significant limitation because they must be physically installed and connected to homes and businesses making them too expensive and time consuming in approximately 90% of the world including areas of deep ocean, large deserts and mountainous areas. Satellite systems solve this digital divide by connecting Mobile Network Operators (MNOs) in remote areas of the world, in-flight connectivity (IFC) in the commercial aviation sector and telemetry for the Internet of Things (IoT) in remote industrial applications like offshore oil fields and maritime shipping (Pantazis et al., 2021). The transition from a relatively unimportant broadcast medium to the backbone network data-transport system is one of the largest in the history of telecommunications technology.

1.3. The Scope and Objectives

To change the old satellite wide beam single-beam architecture to the present High-Throughput Satellites (HTS) is a quantum jump in the design of the payload and spectral use. The transition comes from the broadcasting of one weak signal covering a wide area to hundreds of “spot beams” that are concentrated into a small area. This technological advancement also paves the way for the concept of “Satellite Zooming” the ability to provision large volumes of capacity to specific geographic regions. For the first time, HTS architectures can concentrate electromagnetic energy, removing the previous bandwidth limitation and enabling high resolution data to be collected for pioneering Earth observation, synthetic aperture radar (SAR) imaging and real-time high-definition video transmission (Fenech et al., 2020). The widespread connectivity of our times, it is crucial to understand the intricate technicalities, challenges and future evolution of satellite data communication.

2. Basic Principles of satellite Data Communication

Technical Principles of Satellite Datacom.

The rules of orbital mechanics, the radio-frequency spectrum strategy and the design of digital modulation techniques are all dependent on the usefulness of satellite data communication.

These three pillars are the physical capability, capacity in terms of bit-rate and the ultimate boundedness of any satellite architecture.

2.1. Analysis of Orbital Mechanics

The coverage area, propagation delay and the amount of tracking complexity on earth a satellite can have is dependent on the spatial location of the satellite. Satellites can be placed in the following three orbit regimes above the earth equator: Low Earth Orbit (LEO), Medium Earth Orbit (MEO) and Geostationary Orbit (GEO).

2.1.1. The dynamics and design of LEO Constellation

Low Earth Orbit (LEO) satellites are satellites that orbit at altitudes between 160Km and 2000Km. LEO systems have much shorter propagation delays (5-20 milliseconds) which are very close to the propagation delays of terrestrial fiber systems. This low latency has prompted a renewed interest in LEOs for commercial services with SpaceX's Star link and Eutelsat's One Web being leading providers. But there are a lot of system complexities to LEO architectures. The satellites are not stationary with respect to the Earth but orbit the Earth rapidly, taking about 90 to 120 minutes to complete an orbit. This demands continuous handover protocol and procedure between neighboring satellite and ground antennas to maintain continuous data sessions (Richardia & Westbrook, 2021). Also, hundreds or thousands of satellites would need to be launched into LEO to cover the entire globe in order to keep the system going and this would require a significant investment (CAPEX) and a high number of potential future orbital debris. The Walker delta geometries or polar orbital planes of LEO designs are complicated and are developed to provide minimum coverage gaps at different latitudes for a maximum number of orbital planes.

2.1.2. Medium Earth Orbit (MEO) and Navigation Systems

Medium Earth Orbit (MEO) is the area between LEO and GEO and has a latency of 2,000km to 35,786km which suits certain applications more than GEO. The most apparent use of MEO is in the global navigation satellite system (GNSS) such as GLONASS, Galileo and the Global Positioning System (GPS) at an altitude of about 20,000 km. A satellite orbits the earth about once every 12 hours at this altitude. For the wider individual footprint of MEO, the potential use in broadband communication (e.g., O3b Networks) is also under consideration but in this

case the number of satellites required for global coverage is also reduced and still requires mechanisms of tracking on the ground. While the latency of MEO provides a particularly good value for many broadband applications. The delay from 100 to 150 milliseconds is a disadvantage for applications requiring low latency such as financial trading or virtual reality when compared to the LEO alternatives. (Evans et al., 2022)

2.1.3. The Geosynchronous Orbit (GEO) and the Geosynchronous Advantage

Geostationary Orbit (GEO) is a height of 35,786km above the equator. Orbital period of a satellite in GEO is the same as the period of the earth's rotation (about 23 hours, 56 minutes, 4 seconds) which means that the satellite seems stationary above a point on the surface of the earth. The geosynchronous attribute has not required complex mechanical tracking antennas on the ground in the past, that has been the case with broadcast services and fixed satellite services (FSS). The architecture of GEO ground segments is very simple in order to allow mass production of high gain fixed parabolic antennas very cost-effective. The propagation delay between the ground station and the satellite is about 240 milliseconds per hop (uplink and downlink), resulting in poor performance of the latency-sensitive protocols such as Transmission Control Protocol (TCP) and GEO is not suitable for real-time interactive applications. (Akyildiz & Jornet, 2020)

2.2. Frequency Spectrum Management and Allocation

The RF spectrum is a natural resource that is limited and strictly controlled by the International Telecommunication Union (ITU). Satellite communication is based mainly on microwave frequencies which can pass through the Earth's atmosphere with known attenuation levels. The decision on which frequency band to use is an engineering trade-off between the band width, the protection from degradation by the atmosphere and the complexity of the hardware.

2.2.1. C-Band: The Resilient Legacy Spectrum

The first frequency band to be utilized for commercial satellite communication was in the C-band (4 to 8 GHz) and the typical frequency for uplinks were in the 6 GHz band while downlinks were in the 4 GHz band. The main advantage is its very high resistance to bad weather, especially during heavy rainfall, because the relatively high value for the wavelength

(around 7.5cm) is much larger than the normal rain drops and so the scattering effect is small. Satellite systems in the C-band are severely restricted in transmitted power because of the fact that C-band is also widely used for terrestrial microwave point-to-point links. The ITU has set very stringent power flux density restrictions to protect ground-based infrastructure from satellite interference. So, C-band needs large ground antennas (typically 2 m to 5 m in diameter) to obtain acceptable signal-to-noise ratios (SNR) and has a lower overall bandwidth availability (typically 500 MHz), which means that it cannot be used in the modern high throughput broadband applications. (Maral & Bousquet, 2020)

2.2.2. Ku-Band: Bridging Broadcast and Broadband

The Ku-band (12 to 18 GHz) became the frequency of choice for Very Small Aperture Terminals (VSATs) and Direct to Home (DTH) broadcast television. The bandwidth in Ku-band is much larger than C-band and is typically as large as 1 GHz of usable bandwidth, which facilitates higher data rates. Due to the short wavelength (approximately 2.5 cm), the physical size of ground antennas can be very small than 1 m and can also have a high directivity. Rain fade, is more severe for Ku-band signals: the absorption and scattering of electromagnetic waves due to the water droplets. Ku-band systems use adaptive modulation schemes that reduce data rate during precipitation events to assure signal integrity and Uplink Power Control (UPC) which dynamically boosts the power of the transmitted signal when precipitation occurs, to reduce rain fade. (Garg, 2019)

2.2.3. Ka-Band: The Enabler of High-Throughput Satellites

High Throughput Satellites (HTS) are based on the Ka-band spectrum (26.5 to 40 GHz). The Ka-band transition is motivated by the band's large contiguous blocks of spectrum (up to 3.5 GHz in some regions) that provide it with a capability to deliver far more data than ever before (more than 100 Gbps per satellite). It is the shorter wavelength (around 1 centimeter) that enables the use of very small antenna apertures on the satellite and ground terminal with very high antenna gain. The difficulty in the Ka-band is very high atmospheric attenuation especially during heavy rain when attenuation can be greater than 10 db. This demands very advanced cases of link budget engineering, very narrow-beamed spot beams to achieve high gain and advanced Forward Error Correction (FEC) coding. (Rugamer et al., 2021)

2.2.4. Q/V and W-Bands: The Frontier of Millimeter-Wave Communication

Beyond Ka band, Q/V band (33 to 75 GHz) and W band (75 to 110 GHz) are the next areas of satellite spectrum utilization. The millimeter wave frequencies have gigahertz of available bandwidth, which will enable the next generation of terabit per second satellites. The proliferation condition at these frequencies, is very unfriendly. In clear weather, there are very strong absorptions at the baseline with oxygen and water vapor. The Q/V bands are currently mostly being used for “feeder links” to the satellite which is the high-capacity link between the satellite and the terrestrial gateway but not for direct user links as the ground-based infrastructure to compensate for the losses in the atmosphere (high power transmitters and accurate tracking antennas) is too costly for consumer-based deployment. (Rossi & De Sanctis, 2020)

2.3. Signal Modulation, Multiplexing and Coding Schemes

Signal modulation, multiplexing, and coding schemes form the foundational pillars of modern communications engineering. They determine how efficiently and reliably information can be transmitted across any medium whether terrestrial wireless networks, optical fiber or satellite links. Signal modulation, multiplexing and channel coding are inseparable disciplines that together define the capacity, efficiency and reliability of any communications system. From Shannon’s theoretical foundations to the practical deployment of LDPC codes in satellite and 5G standards, the field has progressed steadily toward the theoretical limits of what physics permits. Future developments including the integration of artificial intelligence into adaptive modulation and coding and the exploration of terahertz bands will continue to push these boundaries further. To maximize the spectral efficiency (bits per second per Hertz) within these constrained frequency bands, satellite communication employs highly complex digital modulation schemes that must operate efficiently within the non-linear constraints of satellite power amplifiers.

2.3.1. Evolution from QPSK to Advanced APSK (DVB-S2/S2X)

The second-generation digital video broadcasting - satellite (DVB-S2) and the extension to this standard, DVB-S2X are the most advanced standardizations in this area. The traditional systems used Quadrature Phase Shift Keying (QPSK) or 2 bits per symbol. The modern HTS

systems use Amplitude and Phase Shift Keying (APSK) which uses 16-APSK and 32-APSK. APSK is a new form of Quadrature Amplitude Modulation (QAM) where the constellation points are in concentric rings. It is essential because satellite traveling wave tube amplifiers (TWTA) and solid-state power amplifiers (SSPA) are internally non-linear at or near their saturation point for maximum power efficiency. The distortions in bit errors of the outer corners of a square QAM constellation result from non-linear amplification. APSK reduces Peak-to-Average Power Ratio (PAPR) which results in the amplifier operating closer to saturation without producing excessive intermodulation distortion. (Eroz et al., 2021)

2.3.2. Adaptive Coding and Modulation (ACM) Mechanisms

Traditional systems have a fixed modulation and coding rate to provide for the worst-case atmospheric conditions (e.g., heavy rain) throughout the coverage area. This leaves a lot of wasted spectrum when weather is clear and the channel SNR is large. The inefficiency is addressed with Adaptive Coding and Modulation (ACM) in DVB-S2X. By dynamically changing the modulation (such as changing between QPSK 1/4, 32-APSK 9/10) frame by frame based on real-time Signal-to-Noise Ratio (SNR) feedback from the remote terminal, ACM enables the satellite gateway to ensure that the link is always in the most comfortable operating point. The dynamic optimization ensures that maximum throughput is achieved from the satellite link at all times leading to an improvement in the overall network efficiency by up to 100% over the performance of a fixed modulation scheme which is an essential capability of HTS architectures. (Casini et al., 2020)

2.3.3. Forward Error Correction (FEC) and LDPC/BCH Implementations

Forward Error Correction (FEC) is a technique used to correct bit errors caused by atmospheric noise and interference without having to retransmit, is an important part of satellite systems. DVB-S2/S2X is a combination of two high-performance codes: an outer Bose-Chaudhuri-Hocquenghem (BCH) code and an inner Low-Density Parity-Check (LDPC) code. The BCH code is very good at correcting random burst errors and the LDPC code offers very good

performance very close to the Shannon Limit. The SNR that the receiver can successfully decode LDPC codes with iterative belief propagation algorithms is almost pushed to the limit of signal acquisition, thus enabling higher-order modulations to be used under marginal weather conditions. (Dong & Zhang, 2021)

3. Satellite Zooming and Precision: High-Throughput Architectures

The conceptual and engineering leap from traditional wide beam satellites to High-Throughput Satellites (HTS) is fundamentally predicated on the principle of spatial discrimination, colloquially termed “Satellite Zooming”. This section rigorously examines the technical architectures that enable this precision.

3.1. The Paradigm Shift: From Wide beam to Spot Beam Technology

A traditional GEO satellite typically utilizes a single broad beam covering a massive geographic area such as an entire continent or an oceanic region. The total capacity of such a satellite is strictly limited by the allocated spectrum (e.g., 500 MHz in Ku-band) and the maximum power of the satellite’s traveling wave tube amplifiers. Because this power is spread over millions of square kilometers, the spectral flux density (measured in Watts per square meter per Hertz) at the Earth’s surface is exceptionally low. Consequently, legacy wide beam satellites rarely exceeded 1 to 2 Gbps in total throughput. HTS architectures dismantle this limitation by replacing the single wide beam with hundreds of narrowly focused spot beams, fundamentally altering the geometry of power delivery. (Fenech et al., 2020)

3.2. Technical Exploration of High-Throughput Satellites (HTS)

3.2.1. Payload Architecture and Digital Channelizers

An HTS’s payload is much more complicated than a bent-pipe satellite. Modern HTS payloads use digital channelizers which are used to digitize the entire allocated spectrum after the low-noise amplifier (LNA) on the uplink. The spectrum is sliced into thousands of individual sub-channels after it has been digitized. The dynamic switching of specific uplink frequencies to any downlink spot beam is referred to as “transparent” routing and is enabled by this digital processing. This frees up the hard-wired connectivity of legacy satellites and enables the

operator to dynamically reconfigure the payload in orbit to handle evolving traffic demand, thus effectively implementing a software-defined satellite (SDS). (Schmidt et al., 2022)

3.2.2. Antenna Gain and Spectral Flux Density Improvements

This course introduces techniques for gaining in antennas and spectral flux density. “Satellite Zooming” is based on the physics of antenna theory. The gain of an ARC is inversely proportional to the square of its beam width. This is accomplished by using large, very precise reflectors (usually 3 to 5 meters in diameter on the satellite) with complex feed arrays which narrow the beamwidth to less than one degree from hundreds of degrees. The concentration of the same amount of RF power into a geographic region of 100 to 500 kilometers diameter results in a highly concentrated spectral flux density at the Earth's surface, typically 100 times as high (20 dB or more) than with a wide beam. It opens the door to higher-order modulation schemes (such as 32-APSK or 64-APSK), drastically smaller and lower cost user terminal equipment and increases the aggregate capacity of the satellite by many multiples of 1 Terabit per second (Tbsp.). (Fang & Chen, 2020)

3.3. Frequency Reuse and Spatial Multiplexing Techniques

3.3.1. Multi-Color Reuse Patterns (4-Color, 7-Color, and Beyond)

It is the “aggressive frequency reuse” that unlocks the power of “Satellite Zooming”. Since the spot beams are very directionally focused and spatially separated, the same frequency bands can be reused concurrently for non-adjacent beams without detrimental co-channel interference. The most popular pattern is the “4-color” frequency reuse pattern. The available spectrum is divided up into four sub-bands (shown conceptually here as red, blue, green and yellow). These sub-bands are then assigned in a checkerboard fashion to spot beams, with no two beams having the same sub-band being physically next to each other. If a 4-color reuse pattern is used for example, on 100 spot beams, the effective usable spectrum is increased by a factor of 25. The capacity multiplication is now taking a new turn with advanced HTS now moving towards 7-color and even 16-color reuse. (Vazquez & Perez-Neira, 2020)

3.3.2. Polarization Discrimination in Spot Beams

Satellite systems make use of orthogonal polarization to further gain spectral efficiency. There are two types of transmission in electromagnetic waves: Linear Polarization (Vertical/Horizontal) and Circular Polarization (Right-Hand/Left-Hand Circular). The ability of such a beam is doubled at a stroke by transmitting one signal in vertical polarization and another completely different signal in horizontal polarization in the same geographic spot beam and the same frequency. This is referred to as Cross-Polarization Discrimination (XPD). Atmospheric limitation is the cause of rain drop flattening causing depolarization of the rain signal in the atmosphere, which results in cross polarization interference (CPI) between the vertically polarized transmitter and the horizontally polarized receiver. (Arora & Singh, 2022)

3.4. Targeted Spot Beam Delivery and Dynamic Beam Hopping

3.4.1. Mechanics of Beam Hopping Illumination

The HTS payload design has evolved from the traditional static spot beam to very dynamic design such as Beam hopping, which is the most noticeable of them. In a conventional HTS, all the spot beams are produced by the satellite and are the same power. But traffic is very unevenly distributed; a spot beam over a very dense urban center might have heavy traffic, while a beam over an ocean is mostly not used. This wasted capacity can't be reconfigured in a static architecture. Beam hopping is possible to overcome this by illuminating spot beams one by one instead of all at once. The satellite payload employs a set of highly agile phased-array antennas or fast-switching solid-state power amplifiers (SSPAs), which can "hop" between various geographic locations a single extremely high-power spot beam. (Alonso et al., 2021)

3.4.2. Resource Allocation Algorithms in Beam Hopping

The dwell time (how long the beam is on at a particular location) is tightly coupled with the transmission of data bursts and this is a highly dependent feature on the super-frame structure of the DVB-S2X standard. This produces a very strong "zooming" effect with a very small amount of time where huge capacity is concentrated precisely where it's required, significantly increasing the statistical multiplexing gain. The ground-based algorithms are very complex

computer programs that control the scheduling of these beam hops. These algorithms were originally static or rule-based, but today's versions employ predictive machine learning algorithms which use traffic matrix information from the past to predict future traffic flow, thus allowing the capacity to be "hopped" to a location in advance of the traffic. (Zhang & Chen, 2021)

3.5. High-Resolution Data Acquisition Applications

3.5.1. Earth Observation (EO) and Synthetic Aperture Radar (SAR)

The resulting HTS zooming and beam hopping combination is transforming the acquisition of high-resolution data. Earth Observation (EO) satellites are used to obtain images at sub-50-centimeter spatial resolution with Synthetic Aperture Radar (SAR) and multispectral sensors. The raw data produced by these sensors is huge; typically, terabytes of data for each orbital pass. Traditionally, these satellites would have to store data on board and wait until they went over a specific ground station before it could transmit the data which would cause a delay in time-critical applications such as disaster management. (Shin et al., 2023)

3.5.2. Real-Time Intelligence, Surveillance, and Reconnaissance (ISR)

High-capacity Ka-band links with geostationary relay satellites are now possible with EO satellites by using HTS spot beam technology. GEO satellite has a “zooming” capability which allows the formation of a very narrow spot beam that tracks the LEO satellite throughout its passage across the sky. The continuous high bandwidth feeder link enables the LEO satellite to continuously downlink all its high-resolution data near real time, regardless of whether the satellite is located above the Earth. In the military and intelligence arena, this translates to the ability to relay synthetic aperture radar images of a target region to an analyst’s terminal anywhere in the world within seconds of acquisition changing the game of operational use of remote sensing assets and permitting dynamic retracking of intelligence, surveillance and reconnaissance (ISR) platforms. (Johnson & Davis, 2020)

4. Comparative analysis: Satellite data communication vs terrestrial data communication

To grasp the unique position that satellite data communication holds, it is essential to make a complete comparison of satellite data communication to the terrestrial networks, in this case fiber-optic and cellular structures. Satellite architectures offer several inherent advantages. Satellite architectures have several inherent advantages.

4.1.1. Ubiquitous, all-round coverage and unrestricted mobility

The greatest advantage of a satellite system is that it is non-geographically dependent. Millions of square kilometers of uniform coverage in a large area that can be covered by a constellation of LEO satellites or a single GEO satellite. Satellite communication is the only possible medium for mobility applications in the absence of terrestrial infrastructure. Satellite links are the only way to provide high-speed satellite data communications to maritime shipping, transoceanic commercial aviation and military operations in serious environments. The only way to deliver broadband satellite data communications to maritime shipping, transoceanic commercial aviation, and military operations in austere environments is by satellite. Moreover, due to its broadcast nature, broadcasting the same information to thousands of users all over the world is the same cost as broadcasting the same information to a single user known as multicast efficiency which cannot be achieved with unicast terrestrial network.

4.1.2. Disaster Resilience and Topological Independence

Satellite networks are unequalled when it comes to surviving local disasters on the ground. The impact of destroying a cell phone tower, cutting a fiber channel or causing its destruction due to an earthquake, tsunami or hurricane will be negligible on a satellite communication backbone. It is not possible to destroy the physical connections to a satellite network at ground level. Satellites play a critical role in coordinating emergency responses in the event of terrestrial communications failure and in delivering first responder with first backhaul. (Sullivan et al., 2022)

4.2.1. The throughput asymmetry compared to Fiber-Optics.

At first glance, satellite communication has some significant issues when compared to terrestrial fiber in terms of raw throughput capacity despite the amazing advances made by HTS. Nearly unlimited scalability is possible by installing additional cables; the modern fiber-optic submarine cables use DWDM or Dense Wavelength Division Multiplexing can support the single-cable speed of 20 Tbsp to 40 Tbsp per cable. The best performance of the most advanced HTS satellites is up to 1-2 Tbsp. The 1 Tbsp is delivered, more importantly, to

thousands of users at the same time in a large footprint. This means that the amount of per-user throughput that a congested satellite beam can provide is usually measured in megabits per second, while the typical amount of throughput that a terrestrial fiber can provide is symmetrical gigabits per second. (Cianca et al., 2022)

4.2.2. The Latency Bottleneck in Geostationary Systems

Latency continues to be a huge drawback with GEO systems. As described, GEO physics has a round trip time (RTT) of 240ms which is adequate to greatly hinder the performance of TCP. High latency will result in an artificially small sliding congestion window for TCP and thus a limit on the size of the “pipe” the protocol will try to fill which will have a significant impact on throughput. Though, one of the disadvantages of LEO constellations is the packet loss problem which arises due to inter-satellite handover events, triggering the congestion avoidance algorithms in TCP, which in turn negatively affecting the user experience. (Chen et al., 2021)

4.2.3. CAPEX, OPEX and Terminal Equipment Costs

Capital costs, operating costs and terminal equipment costs are documented in 4.2.3 CAPEX, OPEX, and Terminal Equipment Costs. The economic models for satellite and terrestrial communications are quite different. The cost of manufacture of satellites, launch insurance and deployment of satellites is astronomical, and for a single GEO HTS it's many hundreds of millions of dollars; the operational expenditure (OPEX) is relatively fixed. But the cost of user equipment is still out of balance. A mechanical VSAT terminal, using a high-power BUC is a relatively expensive (~\$10,000) solution, whereas a terrestrial fiber line connection or mobile Ka-band VSAT terminal with a lower power BUC is a relatively inexpensive (~\$1,000) solution. Also, satellite data communication cost is very high as compared to terrestrial broadband and limited in special consumer market, government and enterprise market. (Kumar & Patel, 2021)

5. Satellite data communication challenges in modern age

Although the introduction of state-of-the-art satellite data communication systems offers a wealth of possibilities, it also comes with its difficulties, such as radio physics, orbital mechanics and information security.

5.1. Attenuation and propagation impairments in the atmosphere

5.1.1. ITU-R Propagation Models and Rain Fade Mechanics

The problem of attenuating the signal from the atmosphere is the most consistent encountered by the satellite communicator, particularly at the Ka bands and higher. The propagation model in the ITU-R P.618 is used as the standard to predict the rain attenuation. It is based on an idea known as the “rain rate” (in millimeters per hour) exceeded due to a certain percentage of the year (usually 0.01%). The attenuation depends on the square of the frequency and cube of the diameter of the rain drop. A moderate-sized thunderstorm with a rain rate of 25 mm/hrs. can create an over 15 dB fade to the signal in a Ka band system operating in the 30 GHz band which is sufficient to eliminate the link connection. (Luini and Capsoni 2021)

5.1.2. Scintillation effects and tropospheric scattering

Besides the rain, the signals are also influenced by tropospheric scintillation, which refers to rapid amplitude and phase fluctuations of the signal caused by small-scale variations of the lower-atmosphere refractive index caused by temperature and humidity fluctuations. The scintillation is more evident at low elevation (with more atmosphere in the path) and at higher frequencies (Ku/Ka bands). Although they typically are shallow (1-3 dB), they can occur very frequently (up to 10 Hz) and may cause problems for automatic gain control (AGC) circuits and phase-locked loops (PLL) in the receiver, e.g., cycle slipping and bit errors.

5.1.3. Site Diversity and Uplink Power Control (UPC) Mitigation

For mitigation of Site Diversity and Uplink Power Control (UPC) the following considerations are important: Complex engineering is needed to reduce the impacts of the atmosphere. Uplink Power Control (UPC) adjusts the power sent by the gateway station dynamically, according to the downlink fade that is experienced by the user terminal or vice versa. A higher cost and highly effective technique, which is called Site Diversity. Two ground-stations are separated by a distance (usually 10-20 km). Intense rain cells are very small in size (typically < 5 km across) and it is statistically very unlikely that both stations will receive severe attenuation at the same time. A switching matrix is used to test the quality of the signal at both stations and traffic is dynamically switched through the station with the best sky view. (Rugamer et al., 2021)

5.2.1. TCP/IP inefficiencies in Long Fat Networks (LFNs)

The Internet Protocol (IP) suite was designed for low latency and low error rate terrestrial networks. Satellite links, in particular GEO satellite links are referred to as Long Fat Networks (LFNs) and have a high Bandwidth Delay Product (BDP). Standard TCP has a very slow starting mechanism which can take a very long time to rise up to fill the satellite pipe, in the case of LFNs. Also, when one packet is dropped, it causes a timeout, forcing TCP to greatly cut its congestion window and re-enter slow-start leading to incredibly low throughput utilization. (Caini & Firrincieli, 2020)

5.2.2. Performance Enhancing Proxies (PEPs) and Protocol Spoofing

In order to tackle this, satellite operators are using Proxy Servers of Performance Enhancing Proxies (PEPs). In a PEP the TCP connection is divided at the edge of the satellite network. The client terminal can communicate with the local PEP using standard TCP/IP, with the remote PEP communication with the local PEP over the satellite link using a satellite optimized protocol. Data integrity is ensured on the satellite link by using an optimized ACK spoofing (most of the time) and by using aggressive FEC and selective repeat ARQ (Automatic Repeat request) to keep the local window open by transmitting fake ACKs in the satellite link as soon as they are received. While effective, PEPs go against the end-to-end principle of IP architecture and can be very troublesome with current end-to-end encryption technologies like

TLS 1.3, where the TCP sequence number are linked to the cryptographic grasp, requiring the more complex integration plans. (Lee & Park, 2022)

5.3. Orbital debris the escalating problem

5.3.1. Kessler's Syndrome and cascade collision dynamics

As a result of the commercialization of space especially the introduction of mega-constellations comprised of thousands of LEO satellites, a significant near-Earth environmental issue is now present. Abandoned satellites, spent rocket bodies and parts of rocket components are known as orbital debris (space trash) which is a threat to the entire future. The Kessler Syndrome is the theory that if there is enough material in LEO to cause collisions to become inevitable. The fragmentation leaves thousands of debris particles with high velocities (7-8 km/s) and these will then crash into other satellites leading to an exponential chain reaction which will make LEO inaccessible for generations to come. (Pelton, 2022)

5.3.2. Post Mission Disposal (PMD) and Active Debris Removal (ADR)

This can only be solved by deorbited satellites at the end of life in accordance with post-mission disposal (PMD) procedures which were previously mandated by FCC to be deorbited within 25 years and now have been brought down to 5 years. Still, PMD assumes that the satellite will have adequate propellant and working avionics at the end-of-life which may not be true. This has stimulated research on the subject of Active Debris Removal (ADR) whereby large debris objects are captured and deorbited using a robotic spacecraft equipped with a harpoon, net or laser ablation system. International space law is still in its infancy and its legal structure is complex at the moment with the question of ownership and liability for nonoperational spacecraft and space debris being a key diplomatic issue. (Gomez & Martinez, 2021)

5.4. The Space Segment vulnerable to cybersecurity threats.

5.4.1. Attack Vectors – Jamming, Spoofing, and Meaconing

Satellite communication systems are faced with an increasing cyber security issue. The most prevalent attack is called jamming and this attack is accomplished by an intruder jamming out a commander or data signal at the satellite's uplink frequency by transmitting a stronger jamming noise signal that results in Denial of Service (DoS). The next step in spoofing is to broadcast the same signal that the ground station would broadcast and hence assume control of the satellite; or provide fake data down the satellite's line of sight (for example fake GPS data). Meaconing involves receiving a valid signal and broadcasting it with a delay, causing the receiving units to give false positions and/or to lose their synchronization. The act of spoofing the signals of a legitimate signal by capturing the legitimate signal and sending it out after a delay, causing the receiver to calculate false positions or to desynchronize is known as meaconing (Bhatt et al., 2023).

5.4.2. Encryption Challenges and quantum-resistant cryptography.

Commercial HTSs may not be able to employ heavy cryptography because of their bandwidth margins and processing power but military satellites are able to rely on such techniques. Also, public-key cryptography (RSA, ECC) which is employed to encrypt connections of satellite telemetry and commands is threatened by the coming of quantum computers. Research is currently underway to shift the paradigm of satellite communication security from Quantum Classical Cryptography (QCC) to Post Quantum Cryptography (PQC), a shift to algorithms that are resistant to quantum (QM) factoring attacks, but are much more costly in terms of computation and therefore in terms of low power processors that can be radiation hardened for their use in space payloads.

6. Future Trends: 6G Integration and AI-Driven Autonomy

Satellite data communications' evolution into the future will always be tied to its convergence with terrestrial cell networks and its integration of AI in space operations.

6.1. Non-Terrestrial Networks (NTN) and 3GPP Standardization

6.1.1. Release 17 and 18 NR Enhancements for Space Networks

The telecom industry is moving fast on the standardization of a satellite communication integration in the next 6G ecosystem known as Non-Terrestrial Networks (NTN) driven by the 3rd Generation Partnership Project (3GPP). Satellite communication was in the past only available as a “stand alone” technology using proprietary protocols. The 6G vision is a natively heterogeneous network that will allow a standard unmodified smartphone to smoothly transition between a 5G terrestrial macro-cell, LEO satellite beam and a high-altitude pseudo-satellite (HAPS) without dropping the connection or switching the protocol stack as illustrated in Figure 6 (Khan et al., 2022).

6.1.2. The Doppler shift compensation and timing advance in LEO

In LEO, the two main options are Doppler shift compensation and timing advance. To make this happen, 3GPP Release 17 and Release 18 have made certain changes to the 5G New Radio (NR) standard specifically for the coverage area of satellites. The Doppler shift on the carrier frequency, due to the action of LEO satellites at 7.5 km/s, is very large and cannot be measured with the standard, terrestrial, receivers. Common frequency offsets and pre-compensation techniques are introduced in the standard. In addition, due to the high propagation delay difference between the satellite overhead and the satellite horizon, the uplink synchronization from millions of different user terminals must be greatly enhanced by using a large improvement in the Timing Advance (TA) mechanism. (Choi & Kim, 2023)

6.2. Optical Inter-Satellite Links (OISL) and Space Mesh Networks

Constellations are now adding Optical Inter-Satellite Links (OISL) to route traffic worldwide without need for terrestrial ground gateways, which are constrained by geopolitical regulations

and geographic constraints. Inter-satellite links with capacities exceeding 100Gbps are possible with free space optics (narrow infrared lasers). In such a way it becomes a constellation that is a high-speed mesh network in space. OISLs can be used to reduce the latency delay required to send a data package between New York and London without having to use Europe for transit, and make it harder for anybody to intercept the data from the ground (Baker & Harris, 2020).

6.3. Autonomous Satellite Operations: Analysis of Artificial Intelligence (AI) for autonomous operations for satellites.

6.3.1. Gaussian Processes for Dynamic Resource Allocation

Mega-constellations and dynamic beam-hopping payloads, and complex inter-satellite routing require cognitive power which is beyond the human level. Artificial Intelligence (AI) is becoming an imperative part of the business process. On the payload side, matrices of network traffic, weather radar data (for prediction of rain fade in advance) and terminal demands are constantly monitored, and the optimal beam hopping illumination patterns, frequency allocation and power distribution are optimized on a sub-second basis with the help of Deep Reinforcement Learning (DRL) algorithms. This results in the HTS payload being truly autonomous cognitive radio (Huang et al., 2021).

6.3.2. Machine Learning for Space Situational Awareness (SSA)

It is important to note that AI is crucial at the constellation level for a Space Situational Awareness (SSA). The number of tracking data of millions of objects of debris is too large to process using existing neural networks, which are tracked by radars and telescopes on the ground and in space. If AI algorithms can predict all the myriad factors that are hard to model – such as the effect of solar radiation pressure and changes in atmospheric drag – they can make more accurate predictions about what happens with the orbit. Furthermore, AI systems can have a preliminary orbit modification analysis and perform the needed ΔV pulses, maximizing the fuel efficiency and assuring the safe operating margins (Serra et al., 2023).

7. Conclusion

Satellite architectures are very well developed and the field of advanced data communication has been transformed. This research essay has clearly illustrated that today's satellite systems are not an old-fashioned bent pipe relay but are very complicated multi-terabit data routers in

the sky. Orbital mastery across the LEO, MEO and GEO regimes allows network architects to optimize orbital parameters to fit their applications' requirements in terms of latency, coverage and capacity. Historical bandwidth limitations have been overcome with aggressive use of higher frequency spectrums especially Ka-band and a plethora of new modulation techniques like DVB-S2X, along with the very efficient LDPC coding. The technological miracle of "Satellite Zooming" is at the core of the paradigm change. Satellite operators can now focus a large amount of spectral capacity on a geographical point via the use of High Throughput Satellites (HTS) aggressive multi-color frequency reuse and dynamic beam hopping. This spatial accuracy not only allows for the highest return on spectrum investment but it is the key to a wide range of real-time, high-resolution data acquisition activities in Earth observation, synthetic aperture radar and intelligence. In terms of simple capacity and ultra-low latency for fixed urban infrastructure there can be no doubt that the winner is the ubiquitous reach of satellite communication but terrestrial fiber optic networks will always have the upper hand. The comparative analysis elaborated that satellite and terrestrial networks are basically synergistic in providing the backhaul, redundancy and mobility coverage that cannot be physically provided by terrestrial networks. But both the systems are far from being flawless. The need for engineering mitigations, the decommission of legacy GEO latency, the looming problem of orbital debris and the growing complexity of cyber risk all demand constant, ongoing efforts of space governance and protocol deconstruction, in addition to engineering mitigations and the integration of quantum-resistant cryptographic architectures. In other words, satellite systems are making their way into the 6G ecosystem from the horizon via the Non-Terrestrial Networks (NTN) space-terrestrial service framework developed by 3GPP. The 3GPP Non-Terrestrial Networks (NTN) space-terrestrial service framework is about to cross the historic boundaries between space and terrestrial networks. Gated by the intelligent algorithms of autonomous operations it will be powered by optical inter-satellite links and create a global omnipresent and pervasive data fabric. In general, the evolution of principles of satellite data communication is more than just a matter of telecommunications engineering; it's a prerequisite for achieving equal digital inclusion, global mobility networks and human expansion into the space domain.

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